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UNIVERSITY OF AGRICULTURE, MULTAN

"Rabbi Zidni Ilma" — My Lord, increase me in knowledge

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DEPARTMENT OF DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

Faculty of Computing

PROGRAMMING FUNDAMENTALS · CS 303

Programming Fundamentals

CS-303

Assignment No. 1

Submitted By

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Program	BS Artificial Intelligence
Semester	1st
Section	A
Department	Data Science and Artificial Intelligence
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Lab no. 1

Task 1: Write a program that prints your name, roll number and program on separate lines.

Program:

main.cpp

Run

```cpp
#include <iostream>
#include <string>
using namespace std;
int main() {
    string name, rollno, program;
    cout << "Enter your name: ";
    getline(cin, name);
    cout << "Enter your Roll no: ";
    getline(cin, rollno);
    cout << "Enter your Program: ";
    getline(cin, program);
    cout << "Name: " << name << endl;
    cout << "Roll No: " << rollno << endl;
    cout << "Program: " << program << endl;
    return 0;
}
```

Output:

Output

```
Enter your name: Hassan Shahzad Qureshi
Enter your Roll no: 2477011
Enter your Program: BS AI (Artificial Intelligence)
Name: Hassan Shahzad Qureshi
Roll No: 2477011
Program: BS AI (Artificial Intelligence)
```

Task 2: Print a box of asterisks 5 rows high and 10 columns wide using cout statements.

Program:

main.cpp Run

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "Name: Hassan Shahzad\n";
    cout << "**********\n";
    cout << "**********\n";
    cout << "**********\n";
    cout << "**********\n";
    cout << "**********\n";
    return 0;
}
```

Output:

Output

```
Name: Hassan Shahzad
**********
**********
**********
**********
**********
```

Task 3: Record three different compiler errors and explain what caused each.

Program:

main.cpp Run

```cpp
#include <iostream>
using namespace std;
int main() {
    // Error 1: Missing semicolon
    // cout << "Hello"

    // Error 2: Undeclared identifier
    // count = 10;

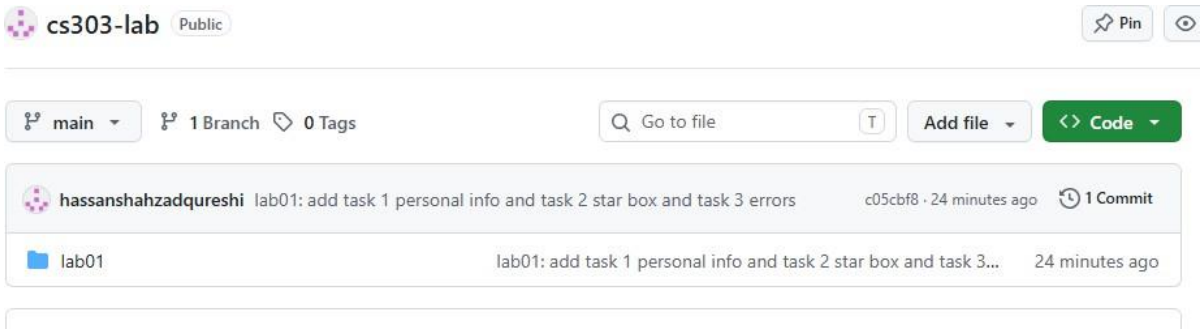
    // Error 3: Missing header or unresolved identifier (e.g., without #include <string>)
    // string s = "test"; // (fails if <string> is omitted in strict standard modes)
    // OR Type mismatch assignment:
    // int x = "Hello";
    cout << "Read comments above for the 3 documented errors.\n";
    return 0;
}
```

Output:

- **Error 1: Missing Semicolon (`expected ';' before '...'`)**
    - *Cause:* Semicolons terminate statements in C++. The parser expects a semicolon before encountering the next statement or closing brace.
- **Error 2: Undeclared Identifier (`'count' was not declared in this scope`)**
    - *Cause:* C++ is statically typed; every identifier must be declared with its type before use.
- **Error 3: Type Incompatible Assignment (`cannot convert 'const char*' to 'int' in initialization`)**
    - *Cause:* Attempting to assign a string literal to a primitive integer variable violates type safety rules.

Task 4: Push your repository to GitHub and share the link.

Link: https://github.com/hassanshahzadqureshi/cs303-lab/



Lab no. 2

Task 1: Read the length and width of a field in metres and print its area in square metres and in kanals (1 kanal = 505.857 m²).

Program:

main.cpp

Run

```cpp
#include <iostream>
#include <iomanip>
using namespace std;
int main() {
    constexpr double kSquareMetresPerKanal = 505.857;
    double length{0.0};
    double width{0.0};
    cout << "Enter field length in metres: ";
    if (!(cin >> length) || length <= 0) {
        cerr << "Error: Invalid length input.\n";
        return 1;
    }
    cout << "Enter field width in metres: ";
    if (!(cin >> width) || width <= 0) {
        cerr << "Error: Invalid width input.\n";
        return 1;
    }
    double areaSqM = length * width;
    double areaKanals = areaSqM / kSquareMetresPerKanal;
    cout << fixed << setprecision(3);
    cout << "\n--- Field Area Results ---\n";
    cout << "Area in Square Metres: " << areaSqM << " m^2\n";
    cout << "Area in Kanals:        " << areaKanals << " kanals\n";
    return 0;
}
```

Output:

Output

```
Enter field length in metres: 123.5
Enter field width in metres: 124.5

--- Field Area Results ---
Area in Square Metres: 15375.750 m^2
Area in Kanals:        30.395 kanals
```

Task 2: Read temperature in Celsius and print it in Fahrenheit and Kelvin with two decimal places.

Program:

main.cpp — Run ▶

```cpp
// Read temperature in Celsius and print it in Fahrenheit and Kelvin with two decimal places.
#include <iomanip>
#include <iostream>
using namespace std;
int main() {
    double celsius{0.0};
    cout << "Enter temperature in Celsius: ";
    if (!(cin >> celsius)) {
        cerr << "Error: Invalid numeric input.\n";
        return 1;
    }
    double fahrenheit = (celsius * 9.0 / 5.0) + 32.0;
    double kelvin = celsius + 273.15;
    cout << fixed << setprecision(2);
    cout << "\n--- Temperature Conversion ---\n";
    cout << "Celsius:    " << celsius << " C\n";
    cout << "Fahrenheit: " << fahrenheit << " F\n";
    cout << "Kelvin:     " << kelvin << " K\n";
    return 0;
}
```

Output:

```
Output

Enter temperature in Celsius: 100

--- Temperature Conversion ---
Celsius:    100.00 C
Fahrenheit: 212.00 F
Kelvin:     373.15 K
```

Task 3: Read a number of seconds and print it as hours, minutes and seconds using / and %.

Program:

main.cpp — Run ▶

```cpp
// Read a number of seconds and print it as hours, minutes and seconds using / and %.
#include <iostream>
using namespace std;
int main() {
    long long totalSeconds{0};
    cout << "Enter total number of seconds: ";
    if (!(cin >> totalSeconds) || totalSeconds < 0) {
        cerr << "Error: Please enter a valid non-negative integer.\n";
        return 1;
    }
    long long hours = totalSeconds / 3600;
    long long remainingSeconds = totalSeconds % 3600;
    long long minutes = remainingSeconds / 60;
    long long seconds = remainingSeconds % 60;
    cout << totalSeconds << " seconds = "
    << hours << " hour(s), "
    << minutes << " minute(s), and "
    << seconds << " second(s).\n";
    return 0;
}
```

Output:

```
Output

Enter total number of seconds: 3665
3665 seconds = 1 hour(s), 1 minute(s), and 5 second(s).
```

Task 4: Demonstrate overflow by adding 1 to INT_MAX and explain the result.

Program:

main.cpp Run

```cpp
// Demonstrate overflow by adding 1 to INT_MAX and explain the result
#include <climits>
#include <iostream>
using namespace std;
int main() {
    int maxVal = INT_MAX;
    cout << "Value of INT_MAX: " << maxVal << "\n";

    // Adding 1 causes signed integer overflow
    int overflowedVal = maxVal + 1;
    cout << "Value of INT_MAX + 1: " << overflowedVal << "\n";
    int underflowedVal = overflowedVal - 1;
    cout << "Value of INT_MAX - 1: " << underflowedVal << "\n";
    return 0;
}
```

Output:

```
Output

Value of INT_MAX: 2147483647
Value of INT_MAX + 1: -2147483648
Value of INT_MAX - 1: 2147483647
```

Lab no. 3

Task 1: Read three numbers and print the largest using nested if.

Program:

task1.cpp

lab03 > task1.cpp

```cpp
// Read three numbers and print the largest using nested if.
#include <iostream>
using namespace std;
int main() {
    double a{0.0}, b{0.0}, c{0.0};
    cout << "Enter three numbers: ";
    if (!(cin >> a >> b >> c)) {
        cerr << "Error: Invalid input.\n";
        return 1;
    }
    double largest{0.0};
    // Using nested if structure
    if (a >= b) {
        if (a >= c) {
            largest = a;
        } else {
            largest = c;
        }
    } else {
        if (b >= c) {
            largest = b;
        } else {
            largest = c;
        }
    }
    cout << "The largest number is: " << largest << "\n";
    return 0;
}
```

Output:

Output

```
Enter three numbers: 10 20 30
The largest number is: 30
```

Task 2: Write a grade calculator for marks 0–100 using the grading scheme of the University semester rules.

Program:

```cpp
// Write a grade calculator for marks 0-100 using the grading scheme of the University semester rules.
#include <iostream>
using namespace std;
int main() {
    double marks{0.0};
    cout << "Enter course marks (0 - 100): ";
    if (!(cin >> marks) || marks < 0.0 || marks > 100.0) {
        cerr << "Error: Invalid marks! Must be between 0 and 100.\n";
        return 1;
    }
    string grade;
    double gpa = 0.0;
    if (marks >= 85.0) {
        grade = "A";   gpa = 4.0;
    } else if (marks >= 80.0) {
        grade = "A-";  gpa = 3.7;
    } else if (marks >= 75.0) {
        grade = "B+";  gpa = 3.3;
    } else if (marks >= 70.0) {
        grade = "B";   gpa = 3.0;
    } else if (marks >= 65.0) {
        grade = "B-";  gpa = 2.7;
    } else if (marks >= 61.0) {
        grade = "C+";  gpa = 2.3;
    } else if (marks >= 58.0) {
        grade = "C";   gpa = 2.0;
    } else if (marks >= 50.0) {
        grade = "D";   gpa = 1.0;
    } else {
        grade = "F";   gpa = 0.0;
    }
    cout << "Marks: " << marks
         << " | Grade: " << grade
         << " | GPA Points: " << gpa << "\n";
    return 0;
}
```

Output:

```
Output

Enter course marks (0 - 100): 90
Marks: 90 | Grade: A | GPA Points: 4
```

Task 3: Write a menu-driven calculator (+, −, ×, ÷) using switch, handling division by zero.

Program:

```cpp
// lab03/task3_calculator.cpp
#include <iostream>
using namespace std;
int main() {
    double num1{0.0}, num2{0.0};
    char op{' '};
    cout << "---------------------------------\n";
    cout << "      Simple Arithmetic Calculator\n";
    cout << "---------------------------------\n";
    cout << "Enter first number: ";
    if (!(cin >> num1)) {
        cerr << "Error: Invalid first number.\n";
        return 1;
    }
    cout << "Enter an operator (+, -, *, /): ";
    cin >> op;
    cout << "Enter second number: ";
    if (!(cin >> num2)) {
        cerr << "Error: Invalid second number.\n";
        return 1;
    }
    switch (op) {
        case '+':
            cout << num1 << " + " << num2 << " = " << (num1 + num2) << "\n";
            break;
        case '-':
            cout << num1 << " - " << num2 << " = " << (num1 - num2) << "\n";
            break;
        case '*':
            cout << num1 << " * " << num2 << " = " << (num1 * num2) << "\n";
            break;
        case '/':
            if (num2 == 0.0) {
                cerr << "Error: Division by zero is undefined!\n";
            } else {
                cout << num1 << " / " << num2 << " = " << (num1 / num2) << "\n";
            }
            break;
        default:
            cerr << "Error: Unsupported operator '" << op << "'.\n";
            break;
    }
    return 0;
}
```

Output:

```
Output

=================================
      Simple Arithmetic Calculator
=================================
Enter first number: 10
Enter an operator (+, -, *, /): +
Enter second number: 10
10 + 10 = 20
```

Task 4: Determine whether a year is a leap year.

Program:

main.cpp

Run

```cpp
// Determine whether a year is a leap year
#include <iostream>
using namespace std;
int main() {
    int year{0};
    cout << "Enter a year: ";
    if(!(cin >> year) || year < 0) {
        cerr << "Error: Invalid Year\n";
    }
    if (year % 4 == 0) {
        cout << year << " is a leap year.\n";
    } else {
        cout << year << " not a leap year.\n";
    }
    return 0;
}
```

Output:

Output

```
Enter a year: 2024
2024 is a leap year.
```

Lab no. 4

Task 1: Print all prime numbers between 1 and 200

Program:

main.cpp Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "Prime numbers between 1 and 200:\n";
    // Loop through numbers from 2 to 200
    for (int i = 2; i <= 200; ++i) {
        bool isPrime = true;
        // Check if current number 'i' has any divisors other than 1 and itself
        for (int j = 2; j * j <= i; ++j) {
            if (i % j == 0) {
                isPrime = false; // Found a factor, so 'i' is not prime
                break;           // Exit inner loop early
            }
        }
        // Print number if it is prime
        if (isPrime) {
            cout << i << " ";
        }
    }
    cout << "\n";
    return 0;
}
```

Output:

Output

```
Prime numbers between 1 and 200:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109
113 127 131 137 139 149 151 157 163 167 173 179 181 191 193 197 199
```

Task 2: Print a right-angled triangle and a pyramid of stars with n rows

Program:

main.cpp Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    int n = 0;
    cout << "Enter the number of rows (n): ";
    cin >> n;
    // 1. Right-angled Triangle Pattern
    cout << "\n--- Right-angled Triangle ---\n";
    for (int i = 1; i <= n; ++i) {
        // Print asterisks equal to the current row number
        for (int j = 1; j <= i; ++j) {
            cout << "* ";
        }
        cout << "\n";
    }
    // 2. Pyramid Pattern
    cout << "\n--- Pyramid ---\n";
    for (int i = 1; i <= n; ++i) {
        // Print leading spaces for alignment
        for (int j = 1; j <= n - i; ++j) {
            cout << " ";
        }
        // Print asterisks to form the pyramid shape
        for (int k = 1; k <= (2 * i - 1); ++k) {
            cout << "*";
        }
        cout << "\n";
    }

    return 0;
}
```

Output:

Output

```
Enter the number of rows (n): 5

--- Right-angled Triangle ---
*
* *
* * *
* * * *
* * * * *

--- Pyramid ---
    *
   ***
  *****
 *******
*********
```

Task 3: Compute the factorial and the sum of digits of a number

Program:

main.cpp Run

```cpp
#include <iostream>
using namespace std;
int main() {
    int num = 0;
    cout << "Enter a positive integer: ";
    cin >> num;
    // Validate input for negative numbers
    if (num < 0) {
        cout << "Factorial is not defined for negative numbers.\n";
        return 1;
    }
    // 1. Calculate Factorial
    long long factorial = 1;
    for (int i = 1; i <= num; ++i) {
        factorial *= i;
    }
    // 2. Calculate Sum of Digits
    int temp = num;
    int sumOfDigits = 0;
    while (temp > 0) {
        sumOfDigits += (temp % 10); // Extract and add the last digit
        temp /= 10;                 // Remove the last digit
    }
    // Display results
    cout << "Factorial of " << num << " = " << factorial << "\n";
    cout << "Sum of digits of " << num << " = " << sumOfDigits << "\n";
    return 0;
}
```

Output:

Output

```
Enter a positive integer: 24
Factorial of 24 = -7835185981329244160
Sum of digits of 24 = 6
```

Task 4: Number-guessing game with at most 7 attempts

Program:

main.cpp

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main() {
    // Seed the random number generator with current time
    srand(static_cast<unsigned int>(time(nullptr)));
    int targetNumber = (rand() % 100) + 1; // Random integer between 1 and 100
    int guess = 0;
    int attempts = 0;
    const int maxAttempts = 7;
    cout << "=== Number Guessing Game ===\n";
    cout << "I have chosen a number between 1 and 100. You have "
         << maxAttempts << " attempts to guess it!\n\n";
    // Loop until user reaches maximum allowed attempts
    while (attempts < maxAttempts) {
        cout << "Attempt " << (attempts + 1) << "/" << maxAttempts << " - Enter your guess: ";
        cin >> guess;
        attempts++;
        // Check user's guess against the target number
        if (guess == targetNumber) {
            cout << "Congratulations! You guessed the correct number in "
                 << attempts << " attempt(s)!\n";
            break; // Exit loop on correct guess
        } else if (guess < targetNumber) {
            cout << "Too low! Try a higher number.\n";
        } else {
            cout << "Too high! Try a lower number.\n";
        }
        // Notify user if max attempts reached without winning
        if (attempts == maxAttempts) {
            cout << "\nGame Over! You have used all attempts. The correct number was "
                 << targetNumber << ".\n";
        }
    }
    return 0;
}
```

Output:

Output

```
=== Number Guessing Game ===
I have chosen a number between 1 and 100. You have 7 attempts to guess it!

Attempt 1/7 - Enter your guess: 15
Too low! Try a higher number.
Attempt 2/7 - Enter your guess: 78
Too high! Try a lower number.
Attempt 3/7 - Enter your guess: 50
Too low! Try a higher number.
Attempt 4/7 - Enter your guess: 67
Too low! Try a higher number.
Attempt 5/7 - Enter your guess: 73
Too high! Try a lower number.
Attempt 6/7 - Enter your guess: 69
Congratulations! You guessed the correct number in 6 attempt(s)!
```

